

AnnotateIQ

AI-Powered Annotation Quality & Operations Intelligence Platform

An end-to-end analytics and automation project for monitoring annotation operations, quality, SLA performance, productivity, anomalies, and project economics. The solution combines Python, Power BI, n8n, and Telegram to turn raw operational records into decision-ready intelligence and scheduled alerts.

DATA QUALITY -> ANALYTICS -> BUSINESS INTELLIGENCE -> AUTOMATION -> ALERTING

5,000 Unique tasks	5,996 QA review records	28 Annotators	5 Projects
92.91% Initial quality score	99.30% SLA compliance	21 Strong anomalies	09:00 Daily alert schedule

PROJECT OBJECTIVE

Build a realistic operations-intelligence system that not only reports what happened, but also identifies abnormal work patterns and automatically delivers actionable alerts to an operations team.

Business problem Annotation programs can look healthy at a high level while hidden bottlenecks, rework, reviewer differences, workload pressure, and cost inefficiencies accumulate underneath.	Solution approach Standardize and enrich operational data, define defensible KPIs, visualize performance in Power BI, detect strong time anomalies, and automate daily notifications through n8n and Telegram.
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Dataset note: The dataset is synthetic and intentionally includes operational patterns and data-quality issues for portfolio demonstration.

Solution Architecture

AnnotateIQ is designed as a lightweight analytics platform rather than a single dashboard. Each layer has a defined role, allowing the project to demonstrate data engineering, analysis, business intelligence, and workflow automation in one coherent system.

END-TO-END ANALYTICS + AUTOMATION PIPELINE



What each layer does

Layer	Primary responsibility	Key output
Data source	Stores raw task, review, annotator, reviewer, project, taxonomy, and automation configuration data.	Operational workbook / CSV feed
Python analytics	Cleans data, resolves duplicates, treats missing values, engineers time and SLA fields, calculates KPIs, and detects anomalies.	Analysis-ready tables
Power BI	Provides executive, trend, quality, operational, and financial views with reusable DAX measures.	Interactive dashboard
n8n	Retrieves the latest CSV feed, filters strong anomalies, aggregates cases, builds alert text in Python, and applies a conditional send rule.	Automated workflow
Telegram	Delivers a concise daily anomaly alert to the operations user.	Actionable notification

Design principles

- Separate raw data from analytical logic so every KPI can be traced back to a defined business rule.
- Use initial QA reviews for quality KPIs to avoid inflating quality with post-rework outcomes.
- Normalize time by input units before labeling a case a strong anomaly.
- Keep automation modular: retrieve -> extract -> filter -> aggregate -> build alert -> condition -> send.

Data Foundation & Cleaning Pipeline

The raw dataset intentionally includes the kinds of inconsistencies that appear in real annotation operations. The Python workflow converts these records into a controlled analytical layer while preserving auditability.

Issue	Treatment	Why it matters
25 duplicate task IDs	Confirmed as exact duplicates except for record ID, then removed.	Prevents double counting in tasks, revenue, labor, and completion metrics.
45 missing time values	Imputed with the median within task_type x complexity; flagged with time_imputed.	Preserves completed records without hiding that an estimate was used.
112 missing completion timestamps	Retained because they correspond to Cancelled, Pending, or In Progress records.	Avoids forcing invalid dates into unfinished work.
Status and language inconsistencies	Trimmed whitespace and standardized status casing.	Creates stable categories for filtering and DAX.
Extreme duration values	Audited using IQR at task_type x complexity level.	Separates legitimate heavy work from suspicious processing time.

Feature engineering

- cycle_time_hours: elapsed time between assignment and completion.
- sla_breached: Boolean SLA evaluation with unfinished work preserved as blank.
- completion_date, completion_month, completion_week: time dimensions for BI analysis.
- time_per_unit: workload-normalized processing time.
- time_outlier and normalized_time_outlier: complementary anomaly signals.
- time_anomaly_type: business-friendly classification of normal vs. unusual cases.

Strong anomaly rule

```
Strong Time Anomaly =
  high total processing time
  AND high normalized time per input unit
```

Result: 21 strong anomalies retained for operational review.

Why use two time signals?

A large task can legitimately take longer. Combining total time with time-per-unit helps avoid flagging normal high-volume work as suspicious.

Auditability

Imputation flags, anomaly categories, and intermediate analytical tables are retained so decisions can be reviewed rather than hidden inside one final metric.

KPI Framework

The dashboard uses business definitions rather than raw column averages. Measures are designed to be interpretable by operations stakeholders and consistent across filters.

98.87% Completion rate	99.30% SLA compliance	9.72 h Average cycle time	92.91% Initial quality
31.96% Rework rate	0.31% Rejection rate	1.98 Errors / initial review	21 Strong anomalies

Core operational and quality measures

KPI	Source table	What it measures	Expected value
Total Tasks	Tasks_Clean	Unique task volume in the current filter context.	5,000
Completion Rate	Tasks_Clean	Completed tasks as a share of workable tasks; cancelled tasks excluded.	98.87%
SLA Compliance	Tasks_Clean	Share of tasks with a valid SLA evaluation that did not breach SLA.	99.30%
Average Cycle Time	Tasks_Clean	Average assignment-to-completion duration.	9.72 hours
Initial Quality Score	Reviews_Raw	Average first-review QA score only.	92.91%
Rework Rate	Reviews_Raw	Share of first reviews requiring rework.	31.96%
Rejection Rate	Reviews_Raw	Share of first reviews rejected outright.	0.31%
Avg. Errors / Initial Review	Reviews_Raw	Mean error count on first review.	1.98

Why initial-review quality matters

If second-round reviews are included in the same average, quality appears better after rework has already corrected failures. AnnotateIQ therefore measures initial quality separately, making the KPI more useful for coaching, staffing, and process diagnostics.

Financial & Project Economics

The financial layer estimates the commercial outcome of each project by combining accepted-task revenue with annotator labor cost. This turns operational performance into an economic view and makes productivity issues visible in monetary terms.

\$1,668.65 Total revenue	\$3,049.32 Estimated labor cost	-\$1,380.67 Estimated profit	-82.74% Overall profit margin
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Project	Client / use case	Revenue	Labor cost	Profit	Margin
PRJ003	SupportMind	\$187.00	\$204.95	-\$17.95	-9.60%
PRJ002	VisionRetail	\$148.86	\$302.44	-\$153.58	-103.17%
PRJ004	HealthText Lab	\$215.95	\$379.34	-\$163.39	-75.66%
PRJ001	VoxBridge	\$500.64	\$846.69	-\$346.05	-69.12%
PRJ005	RoadSense	\$616.20	\$1,315.91	-\$699.71	-113.55%

Calculation logic

- Final QA outcome is taken from the latest review round for each task.
- Revenue is recognized only for tasks finally accepted or accepted with minor issues.
- Labor cost = time_spent_min / 60 x annotator hourly rate.
- Profit = recognized revenue - estimated labor cost.
- Profit margin is calculated only where revenue is greater than zero.

Analytical interpretation The synthetic portfolio is deliberately unprofitable. That creates a realistic management question: which projects, task types, or workflow patterns are consuming more labor than the accepted-task pricing can support?	Decision value The financial page lets a manager move from “quality and throughput” to “where should we redesign pricing, staffing, or process?” without leaving the same BI model.
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Power BI Decision Layer

The Power BI report is organized around four decision levels so that executives, operations leads, and quality teams can use the same model without mixing every metric onto one page.

Page	Purpose	Representative visuals
Executive Overview	Immediate health check across volume, delivery, quality, and financial performance.	KPI cards, task-status donut
Performance Trends	Shows how activity and economics evolve over time and across projects.	Monthly completion trend, profit by project
Quality & Operations	Surfaces rework, rejection, error load, and anomaly concentration.	Quality cards, rework by project, anomalies by annotator
Financial & Project Analysis	Connects revenue, labor cost, and profitability to individual projects.	Revenue vs. labor cost, project margin

DAX design choices

- `DISTINCTCOUNT(task_id)` is used for task volume to protect against duplicate source rows.
- Cancelled tasks are excluded from the completion-rate denominator.
- Blank SLA evaluations are excluded rather than treated as compliant.
- Quality, rework, rejection, and error KPIs use `review_round = 1` for an unbiased first-pass view.
- Financial measures are calculated from the cleaned task-level model so project filters remain interactive.

Example business questions answered

Operations Are completion and SLA performance healthy? Which annotators or batches show unusual timing? Where is rework concentrated?	Management Which projects generate revenue but remain unprofitable? Is cost driven by long task duration, quality rework, or pricing structure?
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Automated Anomaly Monitoring with n8n

The final layer operationalizes the analysis. Instead of requiring a manager to open the dashboard to discover issues, n8n checks the latest feed on a schedule and pushes a concise alert when strong anomalies are present.

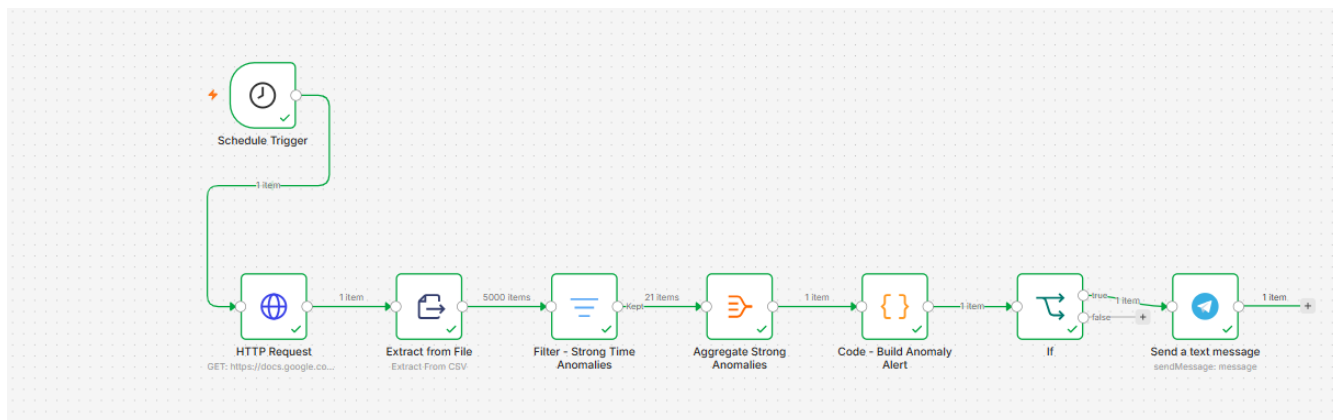


Figure 1. Published AnnotateIQ daily quality-monitor workflow in n8n.

Node	Role
Schedule Trigger	Runs the workflow daily at 09:00 using the Asia/Gaza workflow timezone.
HTTP Request	Retrieves the published CSV data feed.
Extract from File	Parses the CSV into 5,000 task items.
Filter - Strong Time Anomalies	Keeps records classified as Strong Time Anomaly; 21 items in the current dataset.
Aggregate Strong Anomalies	Combines anomaly records into a single list for summarization.
Code - Build Anomaly Alert	Python summarizes counts, affected annotators/projects/batches, and sample cases.
If	Sends the alert only when anomaly_count > 0.
Telegram	Delivers the final message to the configured chat.

Operational Findings & Alert Output

The current synthetic scenario produces 21 strong time anomalies. The automated summary shows that the cases span multiple people and projects, which is more informative than a single outlier count and helps determine whether the pattern is isolated or systemic.

21 Strong anomalies	14 Affected annotators	5 Affected projects	13 Affected batches
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Example Telegram alert

AnnotateIQ - Strong Time Anomaly Alert

Strong anomalies detected: 21
Affected annotators: 14
Affected projects: 5
Affected batches: 13

Sample anomaly cases:

TSK000170 | ANN004 | PRJ003 | PRJ003-B07
TSK000405 | ANN002 | PRJ002 | PRJ002-B01
TSK000435 | ANN002 | PRJ005 | PRJ005-B07

Recommended action: Review the affected tasks, annotators, and batches for unusual processing time.

Embedded scenario signals

- ANN014 has a temporary quality deterioration during 10-24 August.
- Team Beta experiences workload pressure during 15-31 July.
- ANN003 exhibits a speed-quality trade-off pattern.
- Batch PRJ005-B07 is intentionally difficult and contributes multiple anomaly cases.
- RVW004 is configured as a stricter reviewer, creating a useful reviewer-calibration signal.
- PRJ001 improves after 25 August, enabling before/after process analysis.

These embedded patterns make the project useful for interviews: the dashboard is not merely decorative; it contains explainable operational stories that can be investigated from several analytical angles.

Implementation Blueprint

A clean repository structure makes the project easy to review, reproduce, and present. The files below separate source data, notebooks, BI assets, automation, documentation, and screenshots.

```
AnnotateIQ/
  data/
    AnnotateIQ_Dataset_v1.xlsx
    Tasks_Clean.csv
  notebooks/
    AnnotateIQ_Analysis.ipynb
  powerbi/
    AnnotateIQ_Dashboard.pbix
  automation/
    AnnotateIQ_Daily_Quality_Monitor_v1.1.json
  outputs/
    AnnotateIQ_Analytics_Final.xlsx
    KPI_Summary.json
  docs/
    AnnotateIQ_Project_Case_Study.pdf
  screenshots/
    n8n_workflow.png
  README.md
```

Technology stack

Tool	How it is used
Python / pandas / NumPy	Data cleaning, feature engineering, anomaly logic, KPI preparation, alert text generation.
Jupyter Notebook	Transparent, step-by-step analytical development and validation.
Excel / CSV	Portable source and output layer for BI and automation.
Power BI / DAX	Interactive analytical model and stakeholder-facing dashboards.
n8n Cloud	Low-code orchestration, scheduling, conditional logic, and integrations.
Telegram Bot API	Lightweight operational alert delivery.

Portfolio value

- Demonstrates more than dashboarding: the project spans raw-data problems, KPI governance, anomaly detection, economics, BI, and automation.
- Shows practical judgment in metric definitions, especially initial-review quality and SLA denominator handling.
- Connects analytics to action through a working scheduled alert pipeline rather than stopping at visualization.
- Provides a clear interview narrative: problem -> data -> logic -> dashboard -> alert -> business action.

Next Iteration Opportunities

The current version is complete as a portfolio-ready end-to-end system. The following extensions would increase production realism without changing the core architecture.

Priority	Extension	Business benefit
High	Stateful alerting for new anomalies only	Prevents the same historical anomaly set from being re-sent every day.
High	Write alert history to a database or Google Sheet	Creates an audit trail of what was detected, when, and whether it was resolved.
Medium	Dynamic thresholds by task type or project	Makes anomaly rules more adaptive to different operational baselines.
Medium	Reviewer calibration dashboard	Separates genuine annotator issues from reviewer strictness differences.
Medium	Automated daily KPI summary	Combines anomaly alerts with completion, SLA, quality, and rework changes.
Advanced	LLM-generated operations brief	Turns structured KPI changes into a concise management narrative with source-backed values.

Project summary

AnnotateIQ converts synthetic annotation operations data into a complete decision-support workflow. Python establishes a reliable analytical foundation, Power BI translates the model into stakeholder-facing intelligence, and n8n closes the loop by delivering anomaly alerts automatically. The result is a compact but credible example of how data analysis, business intelligence, and automation can work together in an AI operations environment.

END-TO-END VALUE: RAW DATA -> TRUSTED METRICS -> INSIGHT -> AUTOMATED ACTION